Experiment No.: 7

1. Aim: - Design the architecture and implement the autoencoder model for Image Compression.

2. Objective: -

- To design and implement an **autoencoder neural network** for compressing image data.
- To understand how deep learning models can reduce image dimensionality while preserving important features.

3. Outcome:-

- To design and implement an **autoencoder neural network** for compressing image data.
- To understand how deep learning models can reduce image dimensionality while preserving important features.

Software Required: Python

4. Theory

Image compression is the process of reducing the size of an image while maintaining important visual information. It helps reduce storage requirements and improves data transmission efficiency.

An **Autoencoder** is an unsupervised neural network used for feature learning and dimensionality reduction. It consists of two main parts:

- **Encoder:** Compresses the input image into a lower-dimensional representation (latent space).
- **Decoder:** Reconstructs the original image from the compressed representation.

The network is trained to minimize the difference between the input image and the reconstructed image. Autoencoders are widely used in image compression, noise reduction, and feature extraction.

Procedure

Step 1: Import Libraries

Step 2: Load and Preprocess the Data

- Load the MNIST dataset.
- Normalize the image data between 0 and 1.
- Flatten the images into 1D arrays for the autoencoder model.

Step 3: Build the Autoencoder Model

- The encoder reduces the image dimensions (compression).
- The decoder reconstructs the image from the lower-dimensional encoding.

Step 4: Train the Autoencoder

- Train the model with 20 epochs and a batch size of 256.
- Use the training data for both input and target to learn compression and reconstruction.

Step 5: Evaluate the Model

- Use the model to compress and reconstruct test images.
- Plot the original and reconstructed images to evaluate the quality of compression.

5. Results:

Compression Efficiency:

- The autoencoder effectively compresses the images into a lower-dimensional representation of 64 dimensions from the original 784 dimensions.

Reconstruction Quality:

- The reconstructed images retain the main features of the original images, although some details may be slightly blurred due to compression loss.

Loss Reduction:

- The binary cross-entropy loss decreases consistently across the epochs, indicating that the model learns to compress and reconstruct efficiently.

Visualization:

- The reconstructed images closely resemble the original images, demonstrating the model's effectiveness in image compression.

6. Conclusion

The autoencoder model successfully performed image compression on the MNIST dataset, reducing the image dimensions while preserving essential features. The reconstructed images closely resembled the original ones, demonstrating the model's ability to retain visual quality with minimal information loss. These practical highlights the effectiveness of autoencoders for dimensionality reduction and image compression tasks.

7. Viva Question:

1. What is an autoencoder?
2. What are the main components of an autoencoder?
3. What is the function of the encoder and decoder?
4. What is latent space in an autoencoder?
5. What is image compression?
6. How does an autoencoder reduce dimensionality?
7. What is the difference between lossy and lossless compression?

8. References

1. Goodfellow, Y. Bengio, and A. Courville**,** Deep Learning, Cambridge, MA, USA: MIT Press, 2016.
2. F. Chollet**,** Deep Learning with Python**,** New York, NY, USA: Manning Publications, 2018.
3. TensorFlow Tutorials – https://www.tensorflow.org/tutorials
4. Keras Documentation – https://keras.io